NEW ERA PUBLIC SCHOOL

Mid-Term Examination (2026 -27)

Class: X

Subject: Mathematics

Time: 3 Hours **M.M: 80**

General Instructions:

1. This question paper contains - **five sections** A, B, C, D and E. Each section is compulsory. However, there are internal choices in some questions.
2. **Section A** has 18 **MCQs and 02** Assertion-Reason based questions of 1 mark each.
3. **Section B** has 5 **Very Short Answer (VSA)-type** questions of 2 marks each.
4. **Section C** has 6 **Short Answer (SA)-type** questions of 3 marks each.
5. **Section D** has 4 **Long Answer (LA)-type** questions of 5 marks each.
6. **Section E** has 3 **case-based** questions (4 marks each) with sub parts.

Section A

Q1. The number $7 \times 29 \times 23 + 1$ is

(A) a prime number (B) divisible by 23

(C) an odd number (D) a composite number.

Q2. In figure (i), the perimeter of the shaded region is

(A) l (B) $l + a$

(C) $l + 2r$ (D) $l + 2r + a$

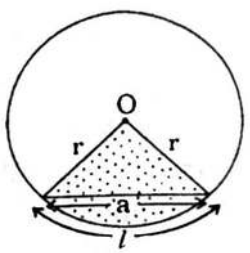


figure (i)

Q3. A car is moving away from the base of a 30 m high tower. The elevation of the top of the tower from the car, at an instant, when the car is $10\sqrt{3}$ m away from the base of the tower is

(A) 30° (B) 45° (C) 60° (D) 90°

Q4. If $a = 2^4 \times 3^3$, $b = 2^3 \times 3^2 \times 5$, $c = 3^n \times 5^2$ and LCM (a, b, c) = $(5^2 \times 3^4 \times 2^4)$, then the value of n is

(A) 1 (B) 2 (C) 3 (D) 4

Q5. If $\tan^2 45° + \cot^2 \alpha = 2$, $0° \leq \alpha \leq 90°$, then the value of α is

(A) 30° (B) 45° (C) 60° (D) 90°

Q6. In figure (ii), if M and N are two points on sides XY and XZ respectively such that MN ∥ YZ, XY = 6 cm and $XN = \frac{1}{4} XZ$, then the length of XM is

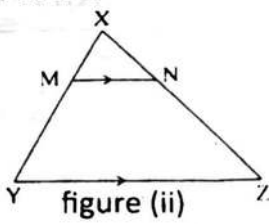


figure (ii)

(A) 1.2 cm (B) 1.5 cm (C) 2 cm (D) 4 cm

Q7. The hour hand of a clock is 7 cm long. The angle swept by it from 7: 00 am to 8: 10 am is

(A) $\left(\frac{35}{4}\right)^o$ (B) $\left(\frac{35}{2}\right)^o$ (C) 35° (D) 70°

Q8. If (k, 3) is the point of intersection of the lines represented by x + py = 6 and x = 15, then (k, p) will be

(A) (15, 3) (B) (15, – 3) (C) (3, 15) (D) (– 15, 3)

Q9. The value of k for which the system of equations kx + 2y = 3 and 2x + y = 5 has a unique solution is

(A) $k \neq 8$ (B) $k = 8$ (C) $k \neq 4$ (D) $k = 4$

Q10. If $x = \sqrt{x}$ is expressed as a quadratic equation in the form $ax^2 + bx + c = 0$, then the value of a + b + c is

(A) 0 (B) 1 (C) 2 (D) 3

Q11. If the HCF of 65 and 117 is expressed in the form of 65m – 117 then the value of m is

(A) 4 (B) 3 (C) 2 (D) 1

Q12. In figure (iii), two concentric circles are shown. The point from which the tangent cannot be drawn to the circle with radius r is

(A) X (B) P (C) Q (D) Y

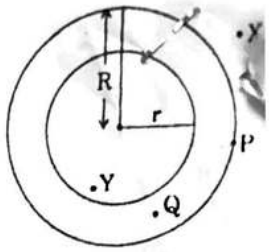


figure (iii)

Q13. If $\sin\theta = \frac{a}{b}$ $(b \neq 0)$ then $\cos\theta$ is given by

(A) $\frac{b}{\sqrt{b^2-a^2}}$ (B) $\frac{b}{a}$ (C) $\frac{\sqrt{b^2-a^2}}{b}$ (D) $\frac{a}{\sqrt{b^2-a^2}}$

Q14. In figure (iv), two similar triangles are shown. The measure of angle P is

(A) 60° (B) 80°

(C) 40° (D) 140°

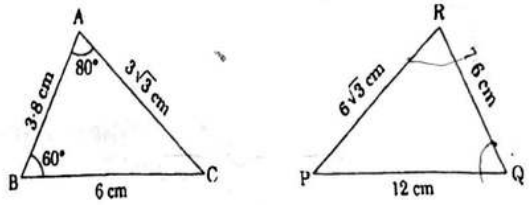


figure (iv)

Q15. The discriminant of the quadratic equation $x^2 - 3x - 2 = 0$ is

(A) 1 (B) 17

(C) $\sqrt{17}$ (D) $-\sqrt{17}$

Q16. The sum and product of the zeroes of the polynomial p(x) are $\left(\frac{-1}{3}\right)$ and 2 respectively. The polynomial p(x) can be

(A) $3x^2 + x + 6$ (B) $x^2 + \frac{1}{3}x - 2$ (C) $3x^2 - x + 2$ (D) $-3x^2 - x + 6$

Q17. If the length of the shadow of a tower is $\sqrt{3}$ times that of its height then the altitude of the Sun at that instant is

(A) 60° (B) 45° (C) 30° (D) 15°

Q18. In figure (v), PT is a tangent to the circle with centre O and radius r. If $\angle POT = 60°$ then the length of OP in terms of 'r' is

(A) $r\sqrt{3}$ (B) $\frac{r}{2}$ (C) 2r (D) r^2

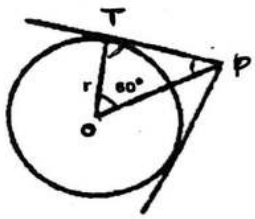


figure (v)

ASSERTION-REASON BASED QUESTIONS

In the following questions, a statement of **Assertion (A)** is followed by a statement of **Reason (R)**. Choose the correct answer out of the following choices.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Q19. **Assertion:** The graph of a polynomial $p(x) = x^2 - x - 6$ intersects the x-axis at $(-2, 0)$ and $(3, 0)$.

Reason: The zeroes of a polynomial $p(x) = x^2 - x - 6$ are (-2) and 3.

Q20. **Assertion:** The tangents drawn at the end points of a diameter of a circle are parallel.

Reason: Diameter is the longest chord of a circle.

Section B

Q21(a). Find the value of k for which the following system of linear equations has no solution:

$3x + y = 1$; $(2k - 1)x + (k - 1)y = 2k + 1$

OR

Q21(b). A man has coins of ₹5 and ₹10 in his purse. If he has total 50 coins that sum up to ₹420, how many coins does he have of each kind?

Q22. A 20 cm long piece of wire is bent to form an arc of a circle of radius $\frac{60}{\pi}$ cm. Find the angle subtended by the arc at the centre

Q23. Prove that $(3\sqrt{2} - 2)$ is an irrational number, given that $\sqrt{2}$ is irrational.

Q24. Solve the following system of equations:
$217x + 131y = 913$; $131x + 217y = 827$.

Q25(a). In figure (v), PQ and PR are tangents to a circle with centre O. If $\angle OPR = 45°$ then prove that ORPQ is a square.

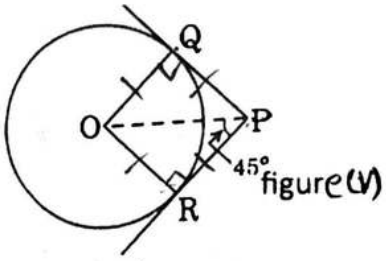


figure (v)

OR

Q25(b). In figure (vi), AB is a diameter to the circle with centre O. BC is the tangent at B. If OP bisects the chord AD and $\angle AOP = 65°$ then find the measure of $\angle ACB$.

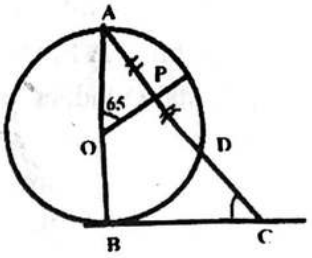


figure (vi)

Section C

Q26. The circumference of a circular field is 360 m. Two cyclists Aslam and Arijit start together from the same point in the same direction at the speed of 10 m/s and 12 m/s respectively around the circular field. After how many minutes will they meet again at the starting point?

Q27(a). Solve the following equation: $\frac{x-1}{x-2} + \frac{x-3}{x-4} = 3\frac{1}{3}$ $(x \neq 2, 4)$

OR

Q27(b). The sum of the squares of two consecutive odd numbers is 394. Find the numbers.

Q28. If α and β are the two zeroes of the polynomial $3x^2 - 4x + 1$ then find the value of $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$.

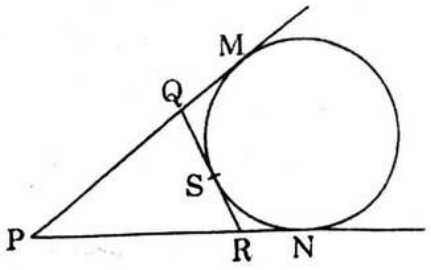


figure (vii)

Q29(a). Prove that a parallelogram circumscribing a circle is a rhombus.

OR

Q29(b). In figure (vii), if a circle touches the side QR of ΔPQR at S and extended sides PQ and PR at M and N respectively then prove that $PM = \frac{1}{2}(PQ + QR + PR)$.

Q30. If $\cos(A + B) = \frac{1}{2}$ and $\tan(A - B) = \frac{1}{\sqrt{3}}$, where $0 \leq A + B \leq 90°$ then find the value of $\sec(2A - 3B)$.

Q31. In figure (viii), $\Delta FEC \cong \Delta GDB$ and $\angle 1 = \angle 2$. Prove that $\Delta ADE \sim \Delta ABC$.

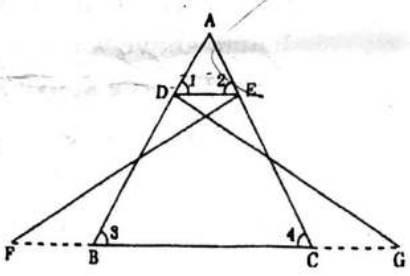


figure (viii)

Section D

Q32(a). In figure (ix), a line AR is drawn through the mid-point Q of side CD of a parallelogram ABCD, which intersects BD at P and produced BC at R. Prove that (i) AQ = QR (ii) PR = 2AP (iii) AP = 2PQ.

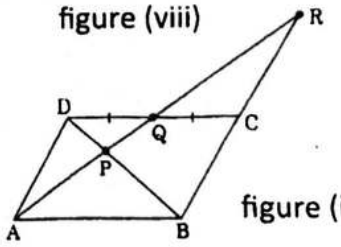


figure (ix)

OR

Q32(b). In figure (x), CD and GH are respectively the medians of ΔABC and ΔEFG. If $\Delta ABC \sim \Delta FEG$, prove that
(i) $\Delta ADC \sim \Delta FHG$ (ii) $\frac{CD}{GH} = \frac{AB}{FE}$ (iii) $\Delta CDB \sim \Delta GHE$.

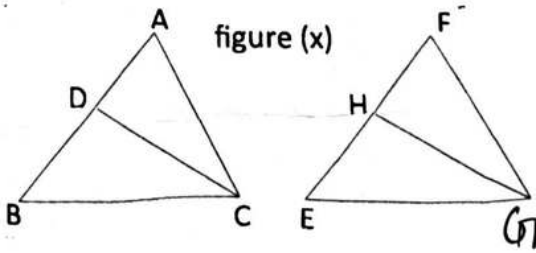


figure (x)

Q33. In a flight of 600 km, an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 200 km/hr and the time of the flight increased by 30 minutes. Find the duration of the flight.

Q34. Solve $2x - y - 2 = 0$; $4x + 3y - 24 = 0$ graphically. Shade the triangular region formed by the given lines and y- axis. Also find the area of the triangle so obtained.

Q35(a) (i) Prove that: $\frac{\tan\theta+\sec\theta-1}{\tan\theta-\sec\theta+1}=\frac{1+\sin\theta}{\cos\theta}$ (ii) If $\cot\theta=\frac{1}{\sqrt{3}}$, find the value of $\frac{1-\cos^2\theta}{2-\sin^2\theta}$.

OR

Q35(b) Evaluate: $\frac{2(\cos^4 60°+\sin^4 30°)-(\tan^2 60°+\cot^2 45°)+3\sec^2 30°}{3\sec^2 60°+2\operatorname{cosec}^2 45°}$

Section E

Q36. An injured bird was found on the roof of a building. The building is 15 m high. A fireman was called to rescue the bird. The fireman used an adjustable ladder to reach the roof. He placed the ladder in such a way that the ladder makes an angle of 60° with the ground.



Based on the above information, answer the following questions.

(i) Find the length of the ladder used by the fireman to reach the top of the building. (1)

(ii) Find the distance of the point on the ground at which the ladder was fixed from the bottom of the building. (1)

(iii) In order to avoid skidding, the fireman placed the ladder, by adjusting its height, in such a way that the bottom of the ladder touches the base of the wall which is opposite to the building, making an angle 30° with the ground.

(a) Draw a neat diagram to represent the above situation and hence find the width of the road between the building and the wall. (2)

OR

(b) Draw a neat diagram to show the above situation and hence find the length of the ladder used by the fireman in this case. (2)

Q37. During a dolphin show, a dolphin jumps out of water travelling at 20 cm/s. Its *height* above water level at t seconds is given by $p(t) = 20t - 16t^2$.



Based on the above information answer the following questions.

(i) Find the zeroes of the polynomial $p(t) = 20t - 16t^2$. (1)

(ii) What will be the height of the dolphin at $t = \frac{3}{2}$ seconds? (1)

(iii) (a) At what time will the dolphin reach a height of 6cm? (2)

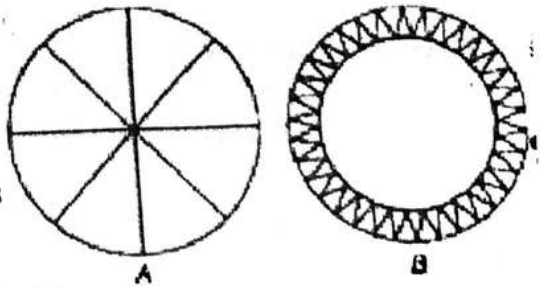
OR

(b) How much distance has the dolphin covered horizontally before hitting the water level again? (2)

Q38. A brooch is a small piece of jwellery which can be used to fasten any dress, blouse or coat.

Brooch A: It is made with silver wire in the form of a circle with diameter 28mm. The wire is also used to make 4 diameters which divide the circle into 8 equal parts.

Brooch B: It is made by using two colored wires silver and gold. The outer part is made with gold. The circumference of silver part is 44 mm and the gold part is 3 mm wide everywhere.



Based on the above information answer the following questions.

(i) Find the central angle of a sector formed by the silver wire in brooch A. (1)

(ii) What is the area of each sector in brooch A? (1)

(iii) (a) Find the area of outer circle in brooch B. (2)

OR

(b) A boy is playing with brooch B. He makes certain revolutions along its edge. How many revolutions he must make to cover 80 π mm? (2)